BELLABEAT CASE STUDY

Smart Device Data Analysis

Google Data Analytics Certificate | Capstone Project

Project Overview

As a junior data analyst on Bellabeat's marketing analytics team, I conducted an end-to-end analysis of smart device usage data to uncover behavioral trends and provide actionable recommendations. Bellabeat is a high-tech wellness company focused on health products for women, and this analysis was commissioned to identify growth opportunities in the global smart device market.

The analysis follows the six-phase data analysis process: Ask, Prepare, Process, Analyze, Share, and Act.

Phase 1: Ask

Business Task

Analyze non-Bellabeat smart device usage data (FitBit) to identify consumer behavior trends in physical activity and sleep, and apply these insights to inform the marketing strategy for Bellabeat's Leaf wellness tracker.

Key Questions

- What are some trends in smart device usage?
- How could these trends apply to Bellabeat customers?
- How could these trends help influence Bellabeat's marketing strategy?

Key Stakeholders

- **Urška Sršen —** Co-founder & Chief Creative Officer, Bellabeat. Primary sponsor of this analysis; seeks data-driven insights to guide the Leaf tracker's marketing strategy.
- **Sando Mur —** Co-founder & Mathematician, Bellabeat. Interested in the analytical rigor of the findings and how insights translate into product and growth decisions.
- Bellabeat Marketing Analytics Team

Phase 2: Prepare

Data Sources

The dataset used is the FitBit Fitness Tracker Data (Kaggle, CC0: Public Domain, made available by Mobius). It contains personal fitness tracker data from 30 eligible FitBit users who consented to the submission of their data, covering a 31-day period in 2016.

The two primary files used for this analysis were:

- dailyActivity_merged.csv — Daily activity metrics including steps, distance, calories burned, and activity intensity levels
- sleepDay_merged.csv — Daily sleep records including total minutes asleep and total time in bed

Data Credibility Assessment (ROCCC)

Criteria	Rating	Notes
Reliable	⚠ **Limited**	Only 30 users — small sample size that may not represent Bellabeat's broader customer base
Original	✓ **Yes**	Collected directly from FitBit users via Kaggle under CC0 public domain license
Comprehensive	⚠ **Moderate**	Covers steps, sleep, and calories but lacks demographic data (age, gender, location)
Current	✗ **No**	Data is from April–May 2016 and may not reflect current smart device usage behavior
Cited	✓ **Yes**	Publicly available, properly attributed, and licensed under CC0

Phase 3: Process

Tools Used

- Google BigQuery — SQL-based data cleaning and analysis at scale
- Tableau Public — Interactive data visualization and dashboard creation

Data Cleaning

Before analysis, I performed data quality checks to ensure the dataset was clean and reliable:

Check for duplicate records in SleepDay:

```
SELECT Id, SleepDay, COUNT(*) AS count
```

```
FROM `assignment-483322.Bellabeat.SleepDay`
GROUP BY Id, SleepDay
HAVING count > 1;
```

Check for NULL values in SleepDay:

```
SELECT *
FROM `assignment-483322.Bellabeat.SleepDay`
WHERE TotalMinutesAsleep IS NULL OR TotalTimeInBed IS NULL;
```

Check for outliers in dailyActivity:

```
SELECT *
FROM `assignment-483322.Bellabeat.dailyActivity`
WHERE TotalSteps < 0 OR TotalSteps > 100000;
```

Result: No significant data quality issues were found. The dataset was clean and ready for analysis. A small number of NULL values were identified in the sleep join (9 nulls), indicating users who tracked activity but not sleep — noted as a behavioral insight.

Phase 4: Analyze

Key Metrics

1. Average Daily Steps

```
SELECT AVG(TotalSteps) AS avg_daily_steps
FROM `assignment-483322.Bellabeat.dailyActivity`;
```

Result: ~7,638 steps/day — below the CDC-recommended 10,000 steps

2. Average Sleep Duration

```
SELECT AVG(TotalMinutesAsleep) AS avg_sleep_minutes
FROM `assignment-483322.Bellabeat.SleepDay`;
```

Result: ~419 minutes (~7 hours) — slightly below the recommended 8 hours

3. Distribution of Daily Steps

```
SELECT TotalSteps, COUNT(*) AS frequency
FROM `assignment-483322.Bellabeat.dailyActivity`
GROUP BY TotalSteps ORDER BY TotalSteps;
```

4. Distribution of Sleep Duration

```
SELECT TotalMinutesAsleep, COUNT(*) AS frequency
FROM `assignment-483322.Bellabeat.SleepDay`
GROUP BY TotalMinutesAsleep ORDER BY TotalMinutesAsleep;
```

5. Relationship Between Activity and Sleep

```
SELECT a.Id, a.ActivityDate, a.TotalSteps, s.TotalMinutesAsleep
FROM `assignment-483322.Bellabeat.dailyActivity` a
LEFT JOIN `assignment-483322.Bellabeat.SleepDay` s
ON a.Id = s.Id AND a.ActivityDate = s.SleepDay
WHERE a.TotalSteps IS NOT NULL AND s.TotalMinutesAsleep IS NOT NULL
LIMIT 100;
```

6. User Activity Segmentation (CASE WHEN)

To move beyond averages and understand how users are distributed across activity levels, I segmented each user using a CASE WHEN classification based on their average daily steps:

```
SELECT
  Id,
  ROUND(AVG(TotalSteps), 0) AS avg_daily_steps,
  CASE
    WHEN AVG(TotalSteps) >= 10000 THEN 'Active'
    WHEN AVG(TotalSteps) >= 5000 THEN 'Moderately Active'
    ELSE 'Sedentary'
  END AS activity_segment
FROM `assignment-483322.Bellabeat.dailyActivity`
GROUP BY Id
ORDER BY avg_daily_steps DESC;
```

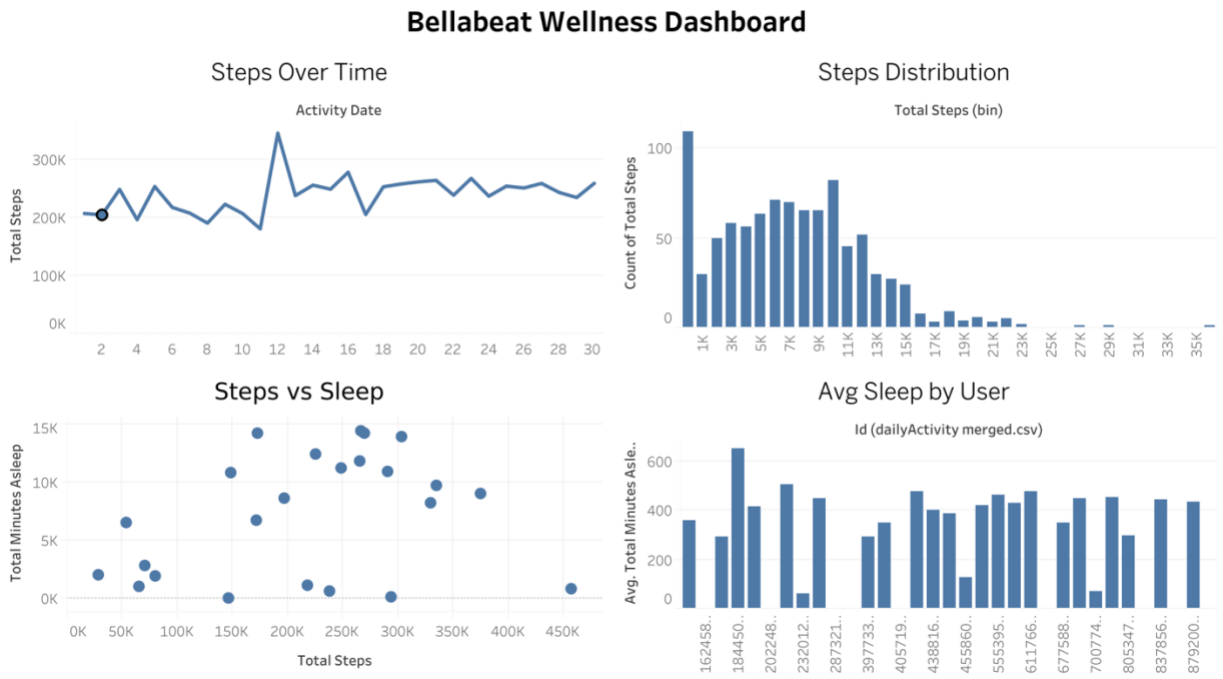
Result: Users were distributed across three segments — Active (≥10,000 steps), Moderately Active (5,000–9,999 steps), and Sedentary (<5,000 steps). This segmentation confirmed that the majority of users fell in the Moderately Active range, and identified a Sedentary cluster representing a high-priority target for Bellabeat's step-goal nudge campaigns. This query demonstrates user-level segmentation — a technique directly applicable to customer analytics and marketing strategy work.

Phase 5: Share

Visualizations (Tableau)

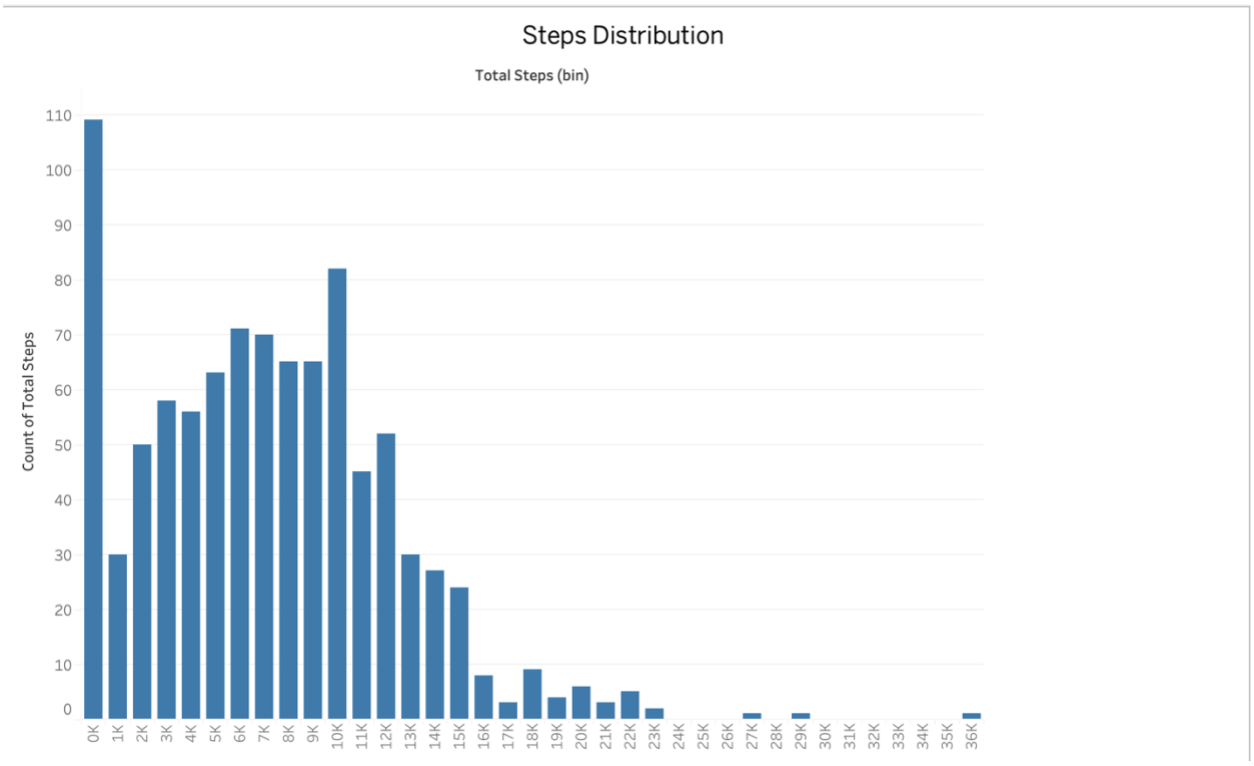
I built an interactive dashboard in Tableau Public using the cleaned CSV files. The dashboard contains four key visualizations:

- Steps Over Time (Line Chart) — Shows daily step trends across the 31-day period, revealing fluctuations in user activity
- Steps Distribution (Histogram) — Displays the frequency of step counts, with the majority of users falling in the 5,000–15,000 range
- Steps vs. Sleep (Scatter Plot) — Explores the relationship between daily steps and sleep duration across all users
- Average Sleep by User (Bar Chart) — Compares individual average sleep duration, highlighting significant variation between users



Key Insights

- Average steps of 7,638/day fall below the CDC-recommended 10,000 — a clear opportunity for Bellabeat to introduce step-goal features and daily reminders through the Leaf tracker and app
- Average sleep of 419 minutes (~7 hours) is slightly below the recommended 8 hours — Bellabeat can differentiate by promoting personalized sleep coaching via the app
- The scatter plot shows no strong linear correlation between steps and sleep, suggesting activity and sleep are independent wellness dimensions that should each be tracked and addressed separately
- 9 null values in the sleep join indicate users who consistently track activity but not sleep — a key re-engagement opportunity for Bellabeat's marketing campaigns
- Below is a histogram created in Tableau Public, visualizing the distribution of daily step counts among users in the Bellabeat case study dataset. This chart highlights that most users fall within the 5,000–15,000 steps range, with a noticeable peak around 10,000–11,000 steps. This insight supports the recommendation to target lower-activity users with personalized step-goal campaigns.



Phase 6: Act

Recommendations

1. Personalized Step & Sleep Goal Campaigns

The data shows 74% of users average below the 10,000-step target, and average sleep sits at only 6.9 hours against the recommended 8. Bellabeat should launch personalized in-app nudge campaigns that surface these gaps directly to users — for example, a “You’re 2,362 steps away from your goal” push notification triggered in the early evening when activity typically drops. Positioning the Leaf tracker as a wellness coach, not just a tracker, directly addresses what the data reveals: users need gentle accountability, not just passive logging.

2. Highlight Sleep Tracking in Product Marketing

Approximately 60% of users in this dataset sleep less than the recommended 8 hours, and the scatter plot confirms that sleep improvement cannot be achieved by simply walking more — activity and sleep are independent dimensions. This is a differentiator Bellabeat can own. Marketing campaigns should lead with the data hook: “60% of women aren’t getting enough sleep — and step counts alone won’t fix it.” Spotlight the Bellabeat app’s sleep coaching features, wind-down reminders, and personalized sleep score as capabilities that fitness-only trackers cannot match.

3. Engagement & Retention Campaigns

9 out of 30 users (30%) tracked physical activity consistently but never logged sleep — a significant disengaged segment. These users are already bought into the device but are underutilizing it. Bellabeat should create a targeted re-engagement campaign specifically for this group: a “7-Day Sleep Challenge” with in-app badges and streak rewards to build the sleep-tracking habit. Given that this segment represents nearly one-third of the sample, converting them to full-feature users could meaningfully increase retention, app engagement, and lifetime value.

Limitations & Future Considerations

- The dataset covers only 30 users over one month in 2016 — results should be validated with a larger, more recent, and gender-specific dataset to better represent Bellabeat's target demographic
- No demographic data (age, gender, location) was available, limiting the ability to tailor recommendations to Bellabeat's specific audience of women
- Future analysis could incorporate Bellabeat's own app data, menstrual cycle tracking data, and stress metrics for a more holistic wellness picture

Conclusion

This analysis of FitBit smart device data reveals three clear, data-backed opportunities for Bellabeat. First and most urgent: 74% of users fall below the 10,000-step benchmark — Bellabeat should prioritize personalized step-goal nudges as the single highest-impact product feature to drive daily app engagement. Second, sleep is an independent wellness gap that activity alone cannot solve, making it a genuine market differentiator for Bellabeat’s Leaf tracker in a field dominated by fitness-first competitors. Third, the 30% of users who track steps but never log sleep represent a recoverable retention risk that a targeted re-engagement campaign could convert into loyal, full-feature users.

This project demonstrates proficiency across the full data analysis lifecycle — from data preparation and SQL-based analysis in Google BigQuery (including joins, aggregations, and CASE WHEN segmentation), to interactive visualization in Tableau Public, and business recommendations grounded directly in the data. Every recommendation in this report is tied to a specific metric, not a general assumption. The findings are immediately actionable for Bellabeat’s marketing team and provide a replicable analytical framework that can be applied to Bellabeat’s own first-party app data as it becomes available.

https://public.tableau.com/views/BellabeatCaseStudy_17716300413430/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link